

UDAL’s Wenlock Hospital Centralized Data Management and Display System

Project Report

1. Project Overview

The UDAL’s Wenlock Hospital Centralized Data Management and Display System is a digital transformation initiative aimed at streamlining internal hospital operations through centralized real-time data display and management. The system connects different departments like the Pharmacy and Blood Bank, enabling automatic synchronization of data and real-time updates on TV displays across the hospital. This project reduces manual effort in updating inventories, increases transparency in stock availability, and enhances efficiency in hospital resource management.

2. Objectives

- To develop a centralized data management system for Wenlock Hospital.
- To automate blood bank inventory updates from Excel to display screens.
- To enable real-time stock availability and updates for the pharmacy department.
- To integrate multiple departmental modules under a single hospital app.
- To implement Kannada transliteration for better accessibility to local staff and patients.
- To ensure the system functions seamlessly in the hospital’s local network environment.

3. System Modules

Pharmacy Module: Displays real-time information on medicine availability. Allows administrators to update stock directly via the connected Excel sheet. Automatic synchronization between the backend and front-end display. Supports Kannada transliteration.

Blood Bank Module: Connects to a live Excel sheet containing current blood unit data. Automatically fetches, processes, and updates blood availability on the display screens. Reduces manual updates and ensures accuracy.

Centralized Application: Acts as the main control hub connecting both modules. Developed as a microservice-based system for modularity. Provides secure access and real-time synchronization across all departments.

4. Technologies Used

Component	Technology / Tool
Frontend	HTML, CSS, JavaScript
Backend	Nodejs
Database / Data Source	Excel (Automated Data Fetching), Local csv
Automation	Google App Script for Excel Automation
Display Integration	TV / Digital Display Screens
Version Control	GitHub

Communication	Google Meet, WhatsApp, In-person
Language Support	English & Kannada (via Transliteration)

5. Key Features

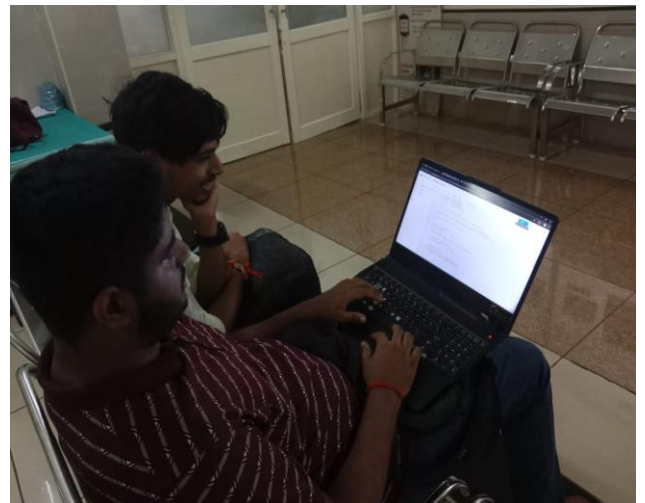
- Real-time synchronization of pharmacy and blood bank data
- Live Blood Bank Display integration with Excel automation
- Pharmacy Display with stock management and Kannada transliteration
- Local network deployment with firewall issue resolution

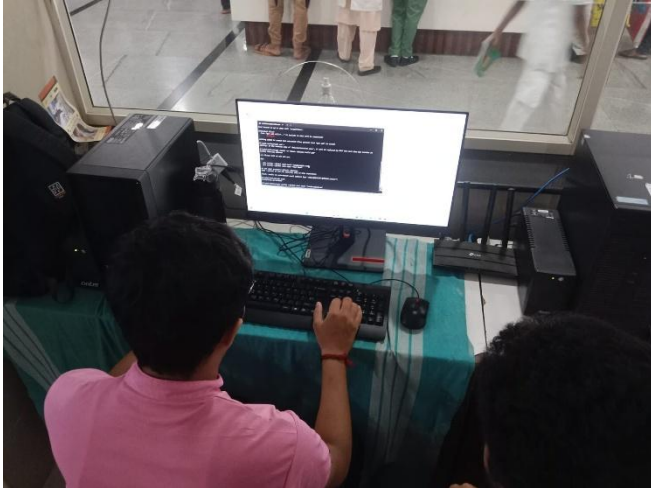
6. Project Development Timeline

Date	Activity / Milestone
August 16	Project officially started. Initial team meeting held to discuss objectives, deliverables, and scope.
August 17	GitHub repository created. Google Meet conducted to finalize workflow, assign roles, and set milestones.
August 18	First visit to Wenlock Hospital; determined display compatibility through trial and error. Post-visit discussion held to document findings.
August 19	First draft of Pharmacy Display and Admin Page completed. Internal review meeting conducted to evaluate progress.
August 20	Structured the Blood Bank Excel Sheet. Short team sync held to define automation approach.
August 21	Automated Excel Sheet to fetch blood bank details locally. Discussion on testing plan and data validation held.
August 22	Meeting to review progress of both modules and assign next set of tasks.
August 24	Kannada transliteration implemented for medicine names. Verification meeting held to check transliteration accuracy.
August 26	Discussion with mentors regarding UI layout and data visibility improvements.
August 28	Blood Bank Excel Sheet connected to Blood Bank Display. Review meeting to confirm smooth synchronization.
September 2	Pharmacy Admin connected to Pharmacy Excel Sheet. Team meeting held to discuss migration issues and solutions.
September 4	Meeting held to plan app integration connecting all microservices.
September 5	Scheduled BOSE Professionals' visit for technical evaluation; pre-visit coordination meeting conducted.
September 8	Initial plan and wireframe for centralized app completed. Feedback session held to finalize interface structure.
September 9	Feedback from Sharath Sir incorporated; BOSE visit completed; Pharmacy Display finalized. Post-review meeting held.
September 10	Discussion on app data flow, login structure, and security features.
September 15	Iteration meeting held to refine app design, assign new subtasks, and track module completion.
September 19	Exams week; short meeting to divide roles and share progress updates.
September 22	Discussion held to plan integration testing and deployment timeline.

Date	Activity / Milestone
September 25	Development of centralized app officially started. Daily stand-up meetings initiated for task updates.
September 28	Internal review of app modules completed; meeting held to resolve UI bugs.
October 2	First draft of the app completed; meeting conducted for feature validation.
October 5	Refinement meeting held to adapt app for Wenlock Hospital's requirements and improve usability.
October 10	Pre-deployment meeting conducted to verify dependencies and firewall permissions.
October 13	On-site work at Wenlock Hospital; refined Blood Bank Excel integration; firewall issues identified. Discussion on troubleshooting approach held.
October 14	Internal meeting to prepare for final deployment and testing.
October 16	App successfully deployed at Wenlock Hospital; firewall issues resolved; functionality confirmed. Final review meeting held to conclude project.

7. Images of work in progress





8. Conclusion

The project successfully enhanced Wenlock Hospital's workflow through digital automation. It provides a scalable model for smart hospital infrastructure under the UDAL initiative, enabling better management, transparency, and accessibility across departments.

